

Project Overview & Proposal: Ambient Pathogen Sentinel (APS)

1. Executive Summary (The Plain-English Concept)

What Are We Building?

Think of this project as "**Smart Smoke Detectors for Pathogens.**"

Traditional public health tracking is slow because it is reactive. When a new virus or bacterial strain emerges, officials only find out after people become symptomatic, visit a clinic, take a PCR test, and wait for lab results—incurring a **7 to 14 day delay**.

Instead of waiting for sick patients to visit clinics, high-traffic communal touchpoints (e.g., automated handwash sinks, greywater traps, towel dispensers) can passively capture biological traces. Our project builds the **AI intelligence layer** that processes these continuous biosensor streams to:

1. Detect whether an observed pathogen carries dangerous new mutations.
2. Backtrack the contagion through the building's physical layout to pinpoint **Ground Zero** (the room or facility where "Patient Zero" introduced it).
3. Automatically generate containment advisories (HVAC airflow changes, localized sanitation directives).

Important Note for the Team: We are **not** working with wet-lab biological samples or physical hardware. The entire project is implemented in **Python** using open-source AI models, network graph libraries, and synthetic outbreak simulations.

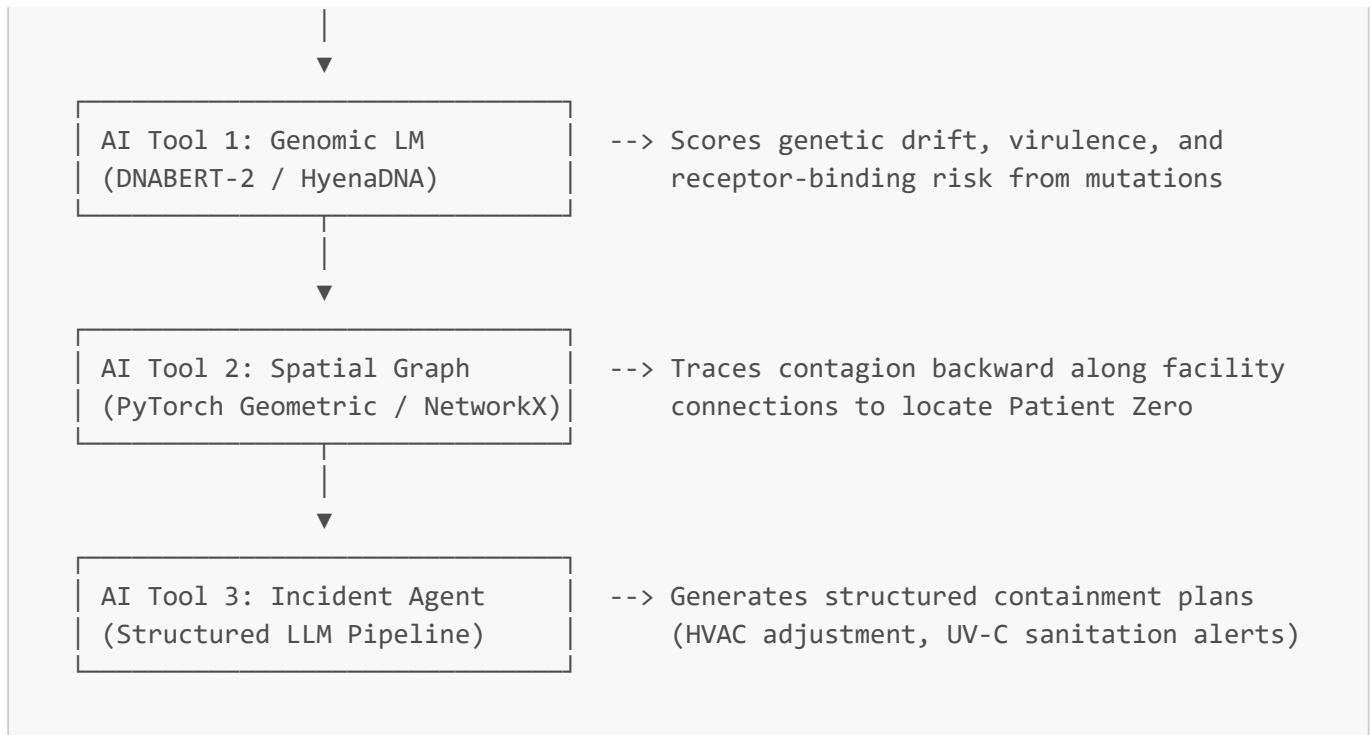
2. Recommended Video Watch List

To build intuition before diving into the code, check out these concepts:

- **Wastewater & Environmental Biosurveillance:**
 - [SciShow: How Sewage Predicts Disease Outbreaks](#) — Demonstrates how environmental sampling detects viral spikes 1–2 weeks before hospital admissions.
- **Biological Foundation Models & Neural Networks:**
 - [3Blue1Brown: But what is a Neural Network?](#) — The foundational visual primer for how weights, activations, and neural network layers process feature streams.
- **Network Graphs & Contagion Dynamics:**
 - [3Blue1Brown: Simulating an Epidemic](#) — Visualizes SIR/SEIR agent transmission, network hubs, and quarantine interventions across dynamic graphs.

3. System Architecture & The 3 AI Stages

[Touchpoint Sensor Percepts]
(Continuous FASTA/k-mer Reads)



1. AI Tool 1 — Mutation & Virulence Scoring:

Treats nucleotide sequences (A, C, G, T) like tokens in a language model. An open pre-trained genomic model reads sequence snippets to determine whether observed variations represent baseline drift or high-risk functional mutations (such as increased receptor-binding affinity or immune evasion).

2. AI Tool 2 — Spatial Back-Tracing (Ground Zero Localization):

Models a facility (e.g., campus building, transit terminal) as a directed graph where nodes are rooms/sensors and edges are hallways, doors, and plumbing lines. When positive readings flag across multiple points over time, an inverse-diffusion search algorithm calculates the most probable origin node.

3. AI Tool 3 — Automated Action Agent:

A constrained language model (using JSON output schemas) parses the graph output and drafts immediate operational directives (e.g., adjust HVAC airflow, deploy targeted UV-C disinfection cycles, issue localized health advisories).

4. Suggested Division of Labor (4 Team Members)

- **Member 1 (Simulation & Environment Design):**
 - Builds the synthetic facility graph using **NetworkX**.
 - Creates the agent-based outbreak simulation (SEIR model) that generates timestamped sensor events.
- **Member 2 (Genomic AI Pipeline):**
 - Loads and configures an open pre-trained genomic model from Hugging Face (**DNABERT-2** or **HyenaDNA**).
 - Implements the script that takes sequence reads and outputs pathogenicity/mutation risk scores.
- **Member 3 (Source Inversion Algorithm):**
 - Implements the graph-traversal / inverse-diffusion algorithm in **PyTorch Geometric** or **NetworkX**.
 - Back-traces infection paths to rank the top candidate nodes for "Patient Zero."
- **Member 4 (LLM Agent, Evaluation & Benchmarking):**

- Builds the structured LLM responder (using Pydantic / JSON mode via Ollama or an open API).
 - Measures accuracy metrics (Top-1 and Top-3 localization rates) and formats final report visualizations.
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5. Course Alignment & Evaluation Plan

- **Use of Existing Tools:** Combines open-source genomic transformers, graph frameworks, and LLM APIs into a unified Python notebook workflow.
- **Measurable Quantitative Evaluation:**
 - Runs 50 Monte Carlo outbreak scenarios across the graph.
 - Measures the **Localization Success Rate** (how often the true seed node is ranked in the top 3 candidates).
 - Benchmarks **Lead-Time Advantage** (how many hours/days earlier detection occurs compared to clinical symptomatic reporting).
- **Failed Experiments Documentation:** Compares simple distance heuristics against graph-based inverse diffusion to document why naive approaches fail (directly fulfilling the rubric).
- **Commercial & Ethical Bonus:** Includes a market justification for transit hubs, hospitals, and enterprise facilities, alongside an analysis of genetic privacy preservation (discarding human DNA to focus strictly on microbial/viral reads).